



Grapevine Culture (Part C): Irrigation and Feeding

Lesson Aim

Specify the techniques used in the culture of grape vines (irrigation and feeding).

Irrigating and Feeding Grapes

Irrigation and feeding are intricately linked because both adequate fertility and adequate irrigation water are necessary for yield quality and quantity. Irrigation and fertility should be balanced to promote crop development and prevent resource loss and pollution.

Plants need adequate water to assimilate and utilise nutrients. However, excess nutrients in the soil can contaminate water, leading to nutrient toxicity in plants, and ground-water pollution through leaching.

Irrigation is generally regarded as supplementary to rainfall. Grapes are frequently grown in areas where there can be a deficit between rainfall and the plants' requirements for good growth and fruit production. However, grapes can tolerate limited rainfall and even thrive in these conditions, and grapes are grown in many vineyards around the world without irrigation.

Irrigation is most widely undertaken in arid and semi-arid climates where soils may be quite fertile, but natural rainfall is insufficient for plant requirements. It is also undertaken successfully in areas with other climate characteristics where it can be used to overcome water shortages in dry times, thus extending growing seasons and hence the types of crops that can be grown, or simply increasing the range of crops. It may also act as a safeguard against times of irregular or unreliable rainfall. Irrigation requirements may range from the complete supply of a plant's needs over the entire growing season down to the occasional small supplementary needs of plants in severe dry seasons. Changes in the global climate, for example in Europe, have led to the use of irrigation where it was previously unnecessary. The use of irrigation to produce improved yields where plants can be grown with just natural rainfall is termed as *supplemental irrigation*.

Excessive Irrigation

Poorly managed, or excessive, irrigation can create problems. Irrigation may even be discouraged in certain regions because of the enhanced risk of disease and loss of fruit quality. Overhead, or sprinkler, irrigation systems, which regularly spray the upper parts of the vine, are more likely to encourage the spread of foliar pathogens than drip systems which only emit water at the base of the vine. Regular drenching of leaves, stems and fruit allows disease-causing organisms, such as bacteria and fungi, to reproduce more efficiently. Many pests must reach an infestation threshold level before they are able to cause damage to the plant. Often it is only when populations rise to a threshold level are they able to cause damage to the plant. In addition to enhancing disease risk, over-generous watering of grape vines will induce excessive growth of shoots. The resulting abundance of vegetative growth will cause shading of lower leaves and grape bunches.

Seasonal Effects of Irrigation

Vine growth slows during winter; at this time, the roots can tolerate high moisture levels in the soil. During the growing season and, in particular leading up to grape harvest, excessive soil moisture can be quite deleterious. High moisture levels result in continuous vigorous growth of vine foliage, resulting in poor quality fruit. Available nutrients are directed to the foliage, preventing proper development of sugars and flavour-forming compounds. Wine produced from such grapes is likely to lack character.

When an over-abundance of water during the harvest period is due to rainfall there is the potential for grape splitting and disease penetration into the fruit. This problem is significant in warm, humid climates.

Symptoms of Water Stress in Grapes

While there are problems with excessive irrigation, water stress at certain periods of the plant's growth cycle will also adversely affect yield and quality. The grape grower needs to be familiar with the tell-tale signs of water stress such as leaf yellowing, withered fruit and shoots, defoliation, and stunted vine growth.

Monitoring and Timing Irrigation

Good irrigation practices can improve the quality and quantity of grape yields. Since grapes consist of 70 – 80% water, irrigation at key times can be beneficial for particular varieties – for example table grapes. Conversely, in the production of wine grapes, Restricted Deficit Irrigation (RDI) may be employed. The vines are kept under controlled stress conditions, to increase grape quality. Irrigation control must be accurate and stringent and only advanced systems of, for example, drip irrigation are suitable for this practice. Effective irrigation, tailored to the particular conditions of the vineyard should, therefore, be a high priority to the viticulturist. Irrigation management needs to be integrated with other management areas, such as pruning (to control fruit loads) and fertility control (particularly nitrogen).

The prime function of an irrigation system is to **prevent** moisture stress, not to relieve moisture stress once it has taken effect. Efficient anticipation of the need for irrigation is the best practice. When to irrigate can be determined by monitoring soil moisture; irrigation is commenced when soil moisture has been depleted to a predetermined amount. Alternatively, grapevine water status can be monitored to determine when the vine is showing a water deficit. Regardless of the method used, once a decision to irrigate has been made, water should be applied to the soil gradually, as early as possible in the season. Large volumes of water applied to the soil close to veraison (the period of the beginning of berry ripening), or close to harvest will adversely affect fruit quality and grapes may fail to reach full maturation and flavour.

Irrigation is essential for juvenile vines in dry periods. Water-stressed young plants will experience significantly retarded growth and, as a result, overall vine size and fruit yield per hectare will be reduced. Healthy young vines will develop good root structure, penetrating deep into the soil and able to better supply water to the plant. Although this will protect the plants from adverse negative effects in times of water stress, irrigation should still be used consistently during the early years to maximise growth potential.

Drip Irrigation

Irrigated vineyards most frequently use drip irrigation systems. Besides being water efficient, drip irrigation systems are useful in vineyards because:

- They minimise the risk of humidity in the vine crown, and so reduce the risk of humidity-related diseases.
- They minimise the wetted area of soil, and so reduce the ability of weeds to compete with vines.
- Nutrients can be supplied with irrigation water, reducing labour and allowing accurate distribution of fertility inputs.

Typically a drip irrigation system will have 13mm diameter black line suspended 15cm above soil level. Drippers, or micro jets, are located adjacent to each vine and not along the intermediate spaces. This set-up provides water precisely where it is required, in the proximity of the root mass. Restricting water supply to other areas along the row will restrict weed growth. Elevating the drip line above soil level will keep it safe from damage e.g. mowing and mechanical weeding. In addition to a primary water filter (depending on the source of the water), in-line filters need be inserted. The quantity of water provided via irrigation will depend on the rainfall amounts, evaporation rates and soil moisture capacity.



Determining the Feasibility of Irrigation

Before undertaking or designing a vineyard irrigation system, a feasibility study should be undertaken to show whether it is desirable to commit resources such as capital, labour, time, and/or water to such a project. Specialist irrigation consultants are often employed for large feasibility studies, as many complex and less obvious factors must be considered, such as topographic features, national and international agricultural and financial markets, transport and communications networks, and state and federal policies on water abstraction.

The economics of smaller vineyard irrigation systems should also be evaluated. Factors to consider include the availability of materials, the expertise and cost required to install and operate the system, and the outcomes and profitability of the system.

At the outset it is important to clearly define the purpose or desired outcome of an irrigation system. The main objective of irrigation systems is to produce a particular desired pattern of plant growth. Maximum vegetative growth doesn't necessarily correspond to maximum yield of the part of the plant desired i.e. fruit. In addition, achieving maximum yield may require inefficient use of available resources, whether it is land, water, equipment or labour. 'Optimum yield' is usually the desired objective. This has been defined as the yield at which the benefit/cost ratio is at maximum, although even this may be hard to achieve if any of the resources required for the irrigation system, e.g. land, water or equipment, is limited.

Design Considerations

Before designing an irrigation system you should consider the following.

Water availability:

- What does it cost?
- Is it available at all times of the year?
- Do you need to build dams or holding tanks to cope with the times of restricted water supply?

Source and quality of water:

- Is it clean?
- Does it carry salt, pollutants, diseases etc.?
- What pressure is it supplied at, or do you need to pump it (if so, how far?).
- Is it from a canal, river, creek, bore, municipal supply or dam?

Regulations:

- Are there any restrictions on use of water?
- Are there any restrictions on building dams, canals etc.?
- Are there any drainage regulations etc.?
- Check with the local council.

Site details:

- You need to know the shape of the site, levels, drainage patterns, locations of buildings, fences, roads, walls etc.
- You need to know the location of plants to be watered.

Finances and labour requirements:

- What money is available to create the system?
- What manpower is available to operate the system?

Water Pressure

The rate at which water is delivered is critical. If water is applied too quickly onto a heavy (e.g. clay) soil, a lot of water may run off the surface and be lost rather than being absorbed into the soil. By choosing the right type of dripper and the appropriate number of emitters for the system you are able to have some control over the rate of application. This can also be controlled by using a pump to increase pressure, or by varying the frequency and length of water application (e.g. pulse watering) to increase water penetration (i.e. giving the water longer periods to soak in).

Vines' Water Requirements

This will vary greatly, depending on the climate and varieties used.

Evapotranspiration

This is water lost from the plant through evaporation and transpiration. Evapotranspiration rates will vary depending upon the vine varieties/cultivars and the climatic conditions.

Soil-Water-Plant Interrelationships

Soil holds various amounts of water, some of which is available to the plants and some that is not. Each soil will vary in its water-holding capacity and drainage characteristics.

Variations in Plantings

Take into account different growth habits, patterns and locations. The way vines are trained and spaced will be a factor. Large-leaved and large vines use more water than smaller ones. Vines trained in the Geneva Double Curtain system use the most water of any system. Early in the season, machine pruned vines develop a greater percentage leaf area and have greater water use than cane-pruned vines. Row spacing will influence water use as will ground cover grown between the rows, such as turf strips.

Soil and Water

Before designing or operating an irrigation system it is important to understand the relationship between soil, water and plants. Soil is made up of a framework of solid material comprising particles of varied sizes and shapes which fit together imperfectly to provide a complex system of pores and channels which provide space within the soil for air and water. When these pore spaces



are filled with water, the soil is saturated. This may occur after irrigation or rainfall. A soil will only remain saturated if it is below the water table and cannot drain freely. The amount of water a soil can hold at saturation depends on the volume of pore space available. This is known as the *saturation capacity* of the soil.

Soil moisture can be grouped into three types of classes: gravitational water, capillary water and hygroscopic water.

Gravitational water is the water that can only remain in the soil for a short time before it drains out under gravity. Gravitational water is found in the larger spaces (pores) between soil particles. This water will drain away from the soil under the influence of gravity, providing there is somewhere for it to drain.

- The flow of gravitational water out of a soil may be inhibited by a barrier such as a hard pan or a high water table.
- The flow of gravitational water out of a soil takes less than one day in sandy soil and three to four days in clay soil.
- Because this type of water disappears from soil quickly, it is not normally included in the water available for use by plants.

A soil is said to be saturated when it contains the maximum amount of gravitational water possible. Saturation capacity is therefore equal to porosity. Porosity is the percentage of the total bulk volume of a soil not occupied by solid particles.

Capillary water is the main source of water for plant growth. This occurs as thin films of water on the soil particles or as droplets in the pore space and is held in place by surface tension. When gravity has removed all the free water a balance is reached where the surface tension binds all the remaining water so that gravity is insufficient to remove it. This condition is known as *field capacity*.

Hygroscopic water is a thin film of water held very firmly to the soil particles so that plants can't remove it. It is not usually available to plants even when the soil is air dry. There are some plants such as xerophytic desert plants that are capable of extracting such water.

Hygroscopic water moves under the influence of gravity as a vapour. (The hygroscopic percentage constant is defined as the moisture content of the soil when it is about air dry. It can only be removed by drying the soil in an oven, if this constant is required for calculations.)

Vines, as is the case with all plants, use water through a process known as *transpiration* whereby the plant acts as a pump, drawing water against the forces holding it in the soil, into the plant roots and transporting it through the stems to the leaves where it is lost to the atmosphere via *evaporation*, an important factor which is influenced by climatic features such as temperature, humidity and wind. If temperature and rate of evaporation are high, then a plant will require more water from the soil than when these conditions are low. Free water is readily utilised by plants, however increasing suction is required to remove the water held by surface tension. When plants reach a stage where they can no longer draw enough water to provide for their needs then the plant may begin to droop. This is known as the *wilting point*. If water becomes available at this stage they will recover, however if water remains unavailable, the plant will reach a point beyond which it will not recover. This is known as *permanent wilting point*. The difference between permanent wilting point and the soil moisture content is known as the *available water* and is thus the water available for plant needs.

The amount of water held in a soil and the amount that is tightly bound will vary from soil to soil. The following table indicates typical soil moisture quantities for different soils expressed as a percentage by weight of dry soil.

Quantities of Soil Moisture in Different Soils

Soil Type	Saturation	Field Capacity	Permanent Wilting Point	Available Water
Fine sand	15-20%	3-6%	1-3%	2-3%
Sandy loam	20-40%	6-14%	3-8%	3-6%
Silt loam	30-50%	12-18%	6-10%	6-8%
Clay loam	40-60%	15-24%	7-10%	8-14%
Clay	40-70%	25-45%	12-20%	13-20%

Symptoms of Water Deficiency

- The first symptom is usually wilting foliage and indicates only a slight water deficiency.
- Over several days, the rate of growth slows.
- The vegetative parts (stems, leaves, shoots) shrivel, die and drop off.

- Photosynthesis is reduced and the vine's ability to produce ripe fruit is reduced.
- The vegetative parts of the vine can begin to take moisture out of the grapes which begin to shrivel.
- Juice volume and berry size decreases.
- Ripening berries may burn in the sun without foliar protection.
- In extreme water deficiency, the plant dies.

Symptoms of Water Excess

- Young plants may become "leggy" (i.e. long, thin stems with elongated internodes - greater spacing between leaves).
- Vines are bushy, vigorous with deep green leaves often with huge berries.
- Plant tissue cells become enlarged and in extreme cases they might burst.
- Berries ripen later and particularly for wine grapes, the juice will be acidic, low in sugar and lacking in flavour.
- In extreme situations leaves become blackened, and the plant may die back or die off completely.

Timing of Irrigations

The zone between wilting point and field capacity is important in irrigation, with the aim being to keep moisture levels within this zone. Remember, the function of irrigation is to **prevent** moisture stress, not to relieve moisture stress once it has taken effect. It is generally accepted that plants take most of their moisture requirements from the upper half of the root zone and consequently only about half of the available water is used. Irrigation is therefore generally required when approximately half of the available water is used up. The volume of water required by the vine is therefore half of the available water in the root zone of the vine when the soil is at field capacity.

Irrigation applications are timed according to how quickly the vines use the available moisture; vines will use more or less water depending on climatic conditions and the availability of nutrients. The rate at which water is supplied by irrigation is also important and is governed by soil infiltration rates (the rate at which water will pass into the soil). If water is supplied at a rate greater than the ground can absorb it, runoff may occur and water will be wasted and lost to the plants. The following table gives an indication of infiltration rates for some soils.

Infiltration rate for different soil types

Soil Type	Infiltration Rate
Coarse sand	2500 mm/hr
Sandy loam	20-100 mm/hr
Loam	10-20 mm/hr
Silty loam	5-10 mm/hr
Clay loam	1-5 mm/hr

The ideal situation is where application rates are equal to infiltration rates. Infiltration rates can also be affected greatly by compaction which causes reduction in pore space and hence space available for water and its passage. Compaction becomes very important when designing irrigation systems for areas that are likely to suffer from a lot of traffic such as heavily used lawn areas. Compacted soils are likely to have infiltration rates as low as 5-10 mm/hr. Measures that may have an effect on increasing infiltration rates such as coring (removing plugs of soil to improve soil aeration) should also be considered when irrigating and application rates adjusted accordingly.

Water Cycling

The best way to wet the soil profoundly is with a cyclic watering method as follows:

- The soil is watered until water begins to run off the surface. At this point the pore spaces near the surface are full of water. Any excess at this point will build up above the surface.
- After leaving the area for a while, excess water in the top layers will soak into the lower layers of the soil, leaving some free pore space again in the top soil. At this point, the area can be watered again until run off point is reached a second time.
- The whole process if repeated several times will continue to cause water penetration to a greater depth in the soil.

Cyclic watering is appropriate in all but sandy soils.

Measuring Water Available to Plants

Calculating Field Capacity of a Vineyard

1. Wet the soil to near saturation. Cover to prevent drying by evaporation. Allow to drain for 2 to 3 days.
2. Take a sample of soil and weigh.
3. Place in an oven at between 100-105°C for 16 hours to dry out. Weigh and record this second weight after drying. (Do not heat at a higher temperature as this can destroy organic material and give a false reading.)
4. Calculate field capacity with the following formula:

$$\text{FIELD CAPACITY} = \frac{\text{Loss in weight}}{\text{Final dry weight}} \times 100$$

Calculating Permanent Wilting Point

1. Fill a pot with the soil to be tested.
2. Plant and grow a plant in the pot until its roots penetrates most of the soil (i.e. roots appear at the bottom of the pot). Use a plant that shows signs of wilting easily when wilting point is reached, e.g. a petunia, tomato, sunflower or fuchsia.
3. Now cease watering until wilting occurs. When the plant wilts, seal the surface of the pot with a sheet of plastic to prevent further loss of water through evaporation.
4. Now place overnight in a humid enclosure (i.e. either a humid greenhouse or plastic tent). If permanent wilting has not occurred the plant will recover. If the wilting persists in the morning, you have then reached permanent wilting point.
5. Now calculate moisture content by drying the soil at 100-105°C for 16 hours, then separate the roots from the soil to find the final dry soil mass.

$$\text{PWP} = \frac{\text{Moist weight} - \text{Dry weight (incl. roots)}}{\text{Final dry weight}} \times 100$$

Available Moisture Range

Available moisture range = Field capacity minus permanent wilting point.

Typical results are as follows:

Available moisture range in different soil types

Soil Type	Field Capacity	Permanent Wilting Point
Clay	45%	30%
Clay loam	40%	25%
Sandy loam	28%	18%
Fine Sand	15%	8%
Sand	8%	4%

In field situations the following factors are used as constants for calculations. The soil is first determined by the feel test.

Available water in different soil types.

Soil Type	Available Water (mm/m of soil)
Clay	165 mm
Clay loam	150
Sandy loam	120
Fine sand	80
Sand	55

Feel test for estimating soil moisture level

How Moist?	What It Feels Like	% Moisture of Field Capacity
Dry	Powdery and dry	0%
Low	Crumbles and doesn't adhere into a ball, even loosely.	Below 25%
Reasonable	Crumbles but will adhere in a ball.	25 to 50%
Good	Adheres into a ball with a little pressure.	50 to 75%
Excellent	Forms a pliable ball which can be rolled into a cylinder.	75 to 100%
Too wet	When squeezed, water drips from the soil.	Over 100% (Over field capacity)

Measuring Air-filled Porosity

Soil porosity can be calculated using the following formula.

$$\text{Air-filled porosity} = \frac{V - M_2/2.65 - (M_1 - M_2)}{V} \times 100 \text{ volume \%}$$

Where: V = volume in cubic centimetres (mL)

M₁ = weight of soil after sampling (g)

M₂ = weight of soil after drying (g) (overnight at 100 to 105 degrees C, or broken up and air dried for a few days)

$$\text{Volume of solids} = \frac{M_2 \text{ (mL)}}{2.65}$$

This is for mineral soils containing only a few percentages of organic matter.

(NB. 1 gram water = 1 mL of water)

$$\text{Air-filled porosity of a potting mix} = \frac{\text{Volume of water drained (mL)}}{\text{Volume of mix (mL)}} \times 100 \text{ (volume \%)}$$

Where the volume of the mix is measured at saturated capacity.

For example, 120 mL of water drains from 600 mL of mix.

Air-filled porosity = 120/600 x 100 = 20 volume %.

That is, 20 % of the volume of the mix was air immediately after it had stopped draining.

(Source: Handreck, K and Black, N, 1986. *Growing Media for Ornamental Plants and Turf*. New South Wales University Press)

Tensiometer

The tensiometer is used to gauge soil moisture characteristics. A field tensiometer can be used to determine when irrigation is required by monitoring the soil moisture content. A vacuum gauge alerts the user as to when watering is needed, using predetermined calculations particular to the soil type. Clay soils may cause problems as they shrink around the tensiometer on drying. The use of tensiometers is restricted to high soil moisture conditions.

Water Extraction by Roots

As a general rule in uniform soil, more root development occurs in the upper levels compared to lower levels. The extraction pattern of moisture from soil by plant roots is generally:

1st quarter depth of roots	40%
2nd quarter depth of roots	30%
3rd quarter depth of roots	20%
Deepest quarter depth of roots	10%

Estimating Water

The amount of water a plant requires is affected by many things, including the following.

Type of Plant

- Some plant varieties utilise more water than others.
- Some plants have a greater resistance to dry conditions (e.g. cacti and succulents are extreme examples).

Rate of Growth

If a plant grows rapidly (perhaps because of its variety, or perhaps due to optimum growing conditions in terms of fertility, climate, etc.), it will use water at a faster rate.

Climate

- In high temperatures soil loses water through evaporation.
- In windy conditions both soil and leaves of the plant lose water faster.
- Higher levels of natural rainfall reduce the need to irrigate. (Make sure you know not only what the annual rainfall is, but also what the distribution of rainfall is throughout the year).

Soil Conditions

- Does the soil drain freely?
- What is the soil's ability to retain water?
- Is there a hard pan or high water table below the soil surface?
- Does the soil repel water when dry (increasing surface run off)?



Available moisture in mm per 100 mm of soil depth

Soil Type	Range	Average (mm)
Clay	22-13	18
Clay loam	20-12	16
Loam	18-10	14
Sandy loam	12-8	10
Sand	8-7	7.5

From the above table you see that a loam soil holds approximately 14 mm of water per 100 mm of depth of soil. Hence: if the plant roots penetrate to a depth of 150 mm, 21 mm of moisture is available.

Drainage In Vineyards

Overly wet ground is undesirable for the following reasons:

- It is unhealthy for the vines. It promotes pest and diseases, reduces oxygen availability to plant roots, and increases humidity levels.
- Wet ground can become inaccessible, slippery or boggy, making it difficult for people and machinery to access the vines and carry out maintenance and harvesting operations.
- Surface water can become stagnant, unhealthy and unpleasant for people working in the vineyard.

To ensure water does not build up in an area you must:

- Prevent water which collects on higher ground from flowing to low spots in the landscape, where it tends to build up.
- Ensure that rainwater moves through the topsoil quickly to the subsoil.
- Remove water from the sub soil quickly, either by passing through to even deeper layers or by collecting it in drains so that it flows away from the site.

Testing Soil Drainage

Soil drainage can be easily tested by observing the way in which water moves through soil which has been placed in a pot and watered. However, when soil is disturbed by digging, its characteristics may change so this may not always be a reliable indication of the soil's drainage properties.

An empty tin can is another way to get a more reliable result. With both the top and bottom removed it forms a parallel-sided tube which can be pushed into the soil to a depth of about 4 cm. Care should be taken to disturb the soil as little as possible. Water can then be poured into the top of the can and allowed to drain down through the soil.

The drainage (or infiltration rate) can then be easily measured in terms of millimetres (mm), or height, of water that soaks into the soil per hour, or more simply the amount the water level drops, measured in millimetres, in the can each hour. This can vary considerably according to the type of soil, from zero infiltration in water-repellent fine sands, up to hundreds of millimetres per hour in coarse sands.

As the soil becomes wetter the infiltration rate decreases until it reaches a constant rate. For some soils this may only take 15 or 20 minutes, for others it may take many hours. If this constant rate is very low, i.e. a few millimetres per hour, then even light rains will cause surface runoff, and water to accumulate in surface depressions.

Vines benefit from good drainage so any vineyard site must have appropriate natural drainage or piping systems must be installed.

An indication of the differences that occur in drainage rates for various soils can be seen from the following figures:

Drainage rates for different soil types (Source: Handreck, K and Black, N, 1986. Growing Media for Ornamental Plants and Turf. New South Wales University Press)

Soil Type	Constant Infiltration Rate (mm per hour)
Deep sands, aggregated silts	more than 20
Deep sandy loams	10-20
Clay loams, shallow sandy loams, soils low in organic matter	5-10
Clays	1-5
Sodic clay soils	less than 1

Improving Subsoil and Surface Drainage

Drainage can be improved in a number of ways. These include:

- Increasing the infiltration rate of water into the soil by improving the permeability of the soil
- Provision of sub-surface drainage to facilitate removal of water from the soil, eg. agricultural pipes
- Provision of surface drainage to remove surface runoff efficiently

Infiltration can be improved by:

- Reducing soil compaction
- Improving soil structure
- Breaking up impermeable layers in the soil such as heavy clays
- Breaking or opening up impermeable surface layers
- Overcoming the water-repelling characteristic of some soils, such as fine sands

Methods that can be used to overcome these problems are cultivation, mixing in soil additives and treating the soil with chemicals.

Cultivation

Surface crusting, impermeable layers and compaction can be rectified to some degree by either machine or hand cultivation. For small jobs, hand cultivators can be used to overcome surface crusting and spades can be used to break up the upper soil layers. For large areas, and problems deeper in the soil profile, machinery is the best option. Deep-ripping cultivators such as mole ploughs can be very effective.

Rotary hoeing of the topsoil is generally ineffective (particularly on silt soils). This method can make soil more workable in the short term, but may increase drainage problems in the long term by causing a hard, impermeable layer below the level of cultivation.

In cases where sub-soil drainage is installed, it may be necessary to undertake special sub-soil cultivations.

Mixing in Soil Additives

These are materials which are mixed with the soil to improve its physical structure with respect to drainage ability.

- Sand - ideally 70% of soil content should be medium or fine sand particles. (Usually added at rates of 250 tonnes per acre or more.)
- Organic matter - peat, compost etc. These materials can be of value mixed with topsoil at the rate of about 25 tonnes per acre or more.
- Other gritty material - occasionally other materials become available which can be used as a sand substitute (e.g. perlite, expanded clay). However most alternative materials are usually too expensive for any but the smallest of garden beds.

Chemical Treatments

These substances are applied to improve the soil's chemical characteristics with respect to drainage ability.

- Clay breaker treatments. Various proprietary treatments are available that disperse clay particles and thus improve the drainage characteristics of clay soils.

- Lime. This white powder does the same job as a clay breaker, but also has side effects - it raises the level of calcium (which can sometimes be a problem) and it makes the soil less acid (i.e. it raises the pH). Unlike clay breaker, lime should be physically dug in to give the best effect.
- Gypsum. This has the same effect as lime but doesn't alter the pH.
- Dolomite lime. This does the same as lime but also adds the nutrient magnesium to the soil, which can be an advantage in some situations, and a disadvantage in others.

*NB: All four treatments above require moist soil conditions over several months to have an effect, so don't expect instant results.

Subsurface Drainage

A subsurface drain is a pipe buried below the ground surface which collects water and moves it away from where it is not wanted. Sub-surface drains are rarely needed in light sandy soils, but may be very important in heavy clay soils.

Layout of Drains

The following patterns are commonly used to lay networks of drainage pipes.

- Herringbone - lateral drains feed from two sides into a central main drain. This system has the advantage of minimising the depth that some of the drains need to be placed. If there is likely to be any subsidence after construction, laterals should be kept at less than 30m in length. On undisturbed ground, up to 90m lengths can be acceptable.
- Grid - laterals run into one main drain from one side. This is less complicated, has fewer junctions, and the main drain can run along the boundary (minimising interference with the main parts of the landscape).

Outlet

Any drain must have a low point where the water drains to and from which it can be disposed of. If it is impossible to find such a low point (i.e. outfall), there are only two possibilities:

- a) Pump the water from the outlet to a suitable disposal point.
- b) Create a soakaway pit - this is a large deep hole filled with porous material such as sand or stones. The water collects in the hole and then slowly disperses into the surrounding soil.

Gradients

A drain should never have a slope less than 1:300, and not less than 1:200 if possible. The slope of a drain should be uniform. Sharp bends should be avoided.

Distance Between Drainage Pipes

In heavy clay, 3 to 5 metres might be necessary. In average soils, 6-9 metre spacings are adequate. In wet spots, it may be necessary to add an additional line and decrease the spacing.

Depth of Drains

There is no firm rule, though 600 mm is usually considered reasonable. If sub-soil cultivation or sand slitting might be carried out at a future date, it is advisable to place drains deeper (e.g. 750 mm).

Characteristics of drainage pipes

Clay pipes - Rigid; should be covered with stones; are generally long term; water soaks through the porous pipe and between joins; joins can sometimes allow soil in, causing blockages.

Corrugated plastic - Flexible; should be covered with stones; water soaks through holes in the pipe; the plastic can break down after a number of years.

Fibreglass strip drain - Flexible; stones are optional; water permeates through outer case; relatively new so long term effects are unknown.

PVC pipe - Rigid; cover with stones; water enters through slits; slippery surface tends to reduce blocking; generally long term.

Laying the Drain

The drain is laid as follows:

1. Dig the trench (must achieve appropriate even slope on base).
2. Lay drainage pipes in bottom of trench.
3. Cover pipes with a coarse material. Suitable materials are hard graded clinker or crushed stone (free of fine dust material). Particle size should be between 10 and 25 mm. This material should fill 30-50% of the trench.
4. Place a layer of sand over the coarse material (to fill within a few centimetres of the top of the trench).
5. Cover the sand with well-draining topsoil.

Surface Drainage

Provision of surface drainage is also important, particularly where you may have large volumes of surface runoff from non-porous surfaces such as concrete, paving or asphalt. This can often be simply achieved by grading surfaces within the landscape so that surface water is directed to a place where it will not cause any harm or it can be efficiently disposed of (for example an opening leading into a sub-surface storm water drain, or to a surface drain). These drains in turn can connect with larger drains or with natural drainage features such as streams.

Surface drains can be simply constructed by creating an elongated dish-shaped depression or trench in the soil which can be left naturally grassed or lined with some material such as asphalt or concrete. Pre-formed concrete, clay or plastic open drains are also readily obtainable. These are generally half circle or rectangular in shape. They can be slotted together or butted up against each other to provide an even slope of sufficient gradient for the free flow of water.

Some Points to Consider When Designing a Drainage System

- Are there any council regulations regarding drainage provisions? For example are there specific drainage easements on the property (or adjacent properties) into which surface or subsurface water can be discharged; do you need permission to connect to existing drainage systems, etc.? Check with your local council or drainage board.
- Will the construction of your vineyard interfere with the drainage patterns of adjoining properties? For example, blocking the natural movement of water from higher properties or directing increased volumes of water into adjoining properties can cause problems.
- Is the drainage system adequate to cope with the amount of water that may result from a heavy rainfall? Your pre-planning information should give you an idea of the nature of rainfall in the area.

Plant Nutrition

Research has indicated that at least 50 different elements may be used by plants. Elements that are required by plants are generally divided into two groups - the major elements (also called macronutrients) and the minor elements (also called micronutrients and trace elements).

Major elements used by plants:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K),
- Magnesium (Mg)
- Calcium (Ca)
- Sulphur (S)

The minor elements include all those elements taken up by plants in only small amounts. The number and importance of elements in this group varies according to the type of plant.

Minor elements used by plants:

- Iron (Fe)
- Zinc (Zn)
- Manganese (Mn)
- Copper (Cu)
- Boron (B)
- Chlorine (Cl)
- Molybdenum (Mo)
- Chlorine (Cl)

Nitrogen, phosphorus, potassium, calcium, magnesium and sulphur are needed by vines in much larger quantities than any other elements (except carbon, oxygen and hydrogen). Most soils have ample supplies of calcium and magnesium; hence, fertilisers which are used in viticulture are usually almost completely made up of nitrogen, phosphorus and potassium foods.

Every nutrient has its purpose, and a deficiency or oversupply of even a minor nutrient can have a major effect on the vine. Deficiencies can be difficult to detect, but as time passes symptoms will appear. Signs are stunted growth, unhealthy leaves that may be mottled, stunted and dying off, distorted stems and undeveloped root systems. If a nutrient is easily dissolved, the older leaves will be affected first, otherwise the growing tips, i.e. the new leaves, will be affected.

Soils high in organic matter hold more nutrients than inorganic soils. Nutrients will be taken up by the vine but also can be lost from the soil through erosion and leaching (the loss of soluble nutrients down through the soil profile in soil water). Compensation for the loss of nutrients by such means needs to be addressed by the viticulturist. Chemical fertilisers will compensate for these losses in rapid manner, however, the use of cover crops between the rows, mineral powders and composts which release nutrients slowly, are all alternatives. These methods require careful planning as benefits will only accumulate gradually.

Choosing the right fertiliser plan helps minimise wastage, reduce costs, and reduce negative effects on the environment, while maximising vine growth and berry quality.

- Nitrogen fertilisers are applied to plants in the order to stimulate green, vegetative growth. Badly timed nitrogen application can have a negative effect on grape quality.
- Adequate phosphorus is essential to maximise root development, for growth and energy transfer. Deficiencies lead to poor fruiting and spindly growth. It should be remembered, however, that of any amount of phosphorus applied to the soil, only about 20% may be immediately available to the plant, the rest being released slowly over a period of time.
- Good levels of potassium are needed for flowering and fruiting. It is also very active in meristematic tissue, where it appears to behave in a similar way to calcium. Deficiency symptoms include marginal chlorosis of older leaves, low yields, weak stems and meristematic necrosis.
- Calcium and magnesium are both required for good fruit development. Stunting and chlorosis are signs of deficiencies, as is a susceptibility to grape rot.

Soil Fertility

Vineyards are frequently established in poor soils, and this can be beneficial for commercial production. Vines are less likely to have excessive growth in poor soils, and some suggest that fruit quality is better.

Grapevine fertility is inextricably related to soil conditions and management. As such, the interplay of plant and soil is a complex one. Grapes are special plants but their fertility requirements for successful growth are not very different from many other agricultural commodities. The favourable rooting characteristics of many types of grapevines allow them to grow well in low fertility soils. They can also grow well in a wide variety of soils. Successful grape production relies heavily on cultural treatment of the grapevine, including variety selection, training, harvest timing and, eventually, wine-making procedures.

Soil Type and the Production of Excellent Wines

The physical characteristics as well as the chemical components have an effect on the grape flavour. French Champagne is best made from grapes grown on solid chalk soils. The alkalinity of these soils makes the wine more acid, a necessary characteristic in Champagne production. The great clarets of Bordeaux grow on poor gravely soils. Fine delicate moselles from Germany grow on broken flakes of slate. Some of the finest ports in the world are grown in the Douro hills on heavy granitic and schistous rock.

Characteristics of good agricultural soils:

- Well drained
- Deep-rooting zone capacity
- Easily penetrated by air, water and roots
- Good water-holding capacity
- Maintains a balanced nutrient supply
- Resists erosion

Changes that can lead to bad soils:

- Reduction in vegetation cover, leading to increased rates of erosion
- Loss of soil structure due to poor cultivation techniques
- Reduction in soil fertility by not replacing lost nutrients
- Increase in salinity
- Waterlogging
- Soil acidification through the extensive use of acidifying fertiliser
- Pollution of soils through the use of persistent or toxic agricultural chemicals

The viticulturist extracts valuable information for soil fertility reports. The basic rule is that if soil fertility is high then it is not advisable to add either synthetic fertilisers or natural compounds such as manures and compost. Not only would the grower be wasting money, but, also risking a decrease in grape quality by added unneeded fertilisers. However, if the soil report indicates deficiencies in nutrient levels it is recommended that they are dealt with specifically. Enhancing those nutritional elements that show low levels is more advantageous, and cost effective, than applying general fertilisers. The result is well balanced soil fertility. Balanced nutrient levels may be of greater benefit than overall high levels.

Vines are frequently fertilised with complete fertilisers in the first few years of vineyard establishment, in order to develop strong vines. The goal is to ensure that important elements are available and those being assimilated by the plant are replaced annually. Subsequent fertility requirements are frequently determined by soil and leaf testing. Some growers will stop adding fertilisers after the initial establishment period and only add compounds if poor growth occurs.

Some vineyards apply compost on a regular basis. While composts do supply some fertility, they also act to improve the soil, and can improve disease resistance and general vineyard health.

Choice of Fertiliser/Timing of Application

The choice and amount of fertiliser applied to a vineyard is dependent on desired yield, variety of grape, soil type, irrigation efficiency and whether or not a leguminous cover crop is grown in the vineyard, between the growing seasons. All sources of nitrogen must be taken into account, such as nitrates (NO_3) from irrigation water or nitrogen from any cover crop. The fertiliser composition must be selected on the basis of the vine's needs. Petiole analysis of nutrient levels within the vine is the favoured tool for this purpose.

The first application of fertiliser is applied at bud break; an additional application of sulphate of potash should be made at the same time. Second and third applications can be made at full bloom and veraison if indicated by petiole analysis. A high nitrogen general purpose fertiliser is suitable for the first application; choice of NPK ratio thereafter is dependent on the indication of petiole analysis.

A leguminous cover crop e.g. lupins or lucerne, can be sown between the rows after harvest. This helps fertilise the soil; and if ploughed back in, the organic material helps the soil structure. A permanent cover crop may sometimes be used in over vigorous vineyards; it will be mown through the summer to prevent seeding.

The soil pH is very important, both for efficient take up of nutrients and for its effect on berry pH. Sufficient lime must be added if required to maintain pH levels at 6.0-6.5. Grape vines often suffer from magnesium deficiency. If petiole analysis indicates this deficiency treat with magnesium sulphate.

Where possible organic mulch spread around the base of each vine will be of great benefit. Avoid contact with the stem itself, however.

Fertigation

Fertigation is the practice of adding soluble fertiliser to irrigation water. Fertigation is widely used in commercial vine production where the vineyard is irrigated. Mineral depletion is accelerated, by leaching, in the area of limited soil volume under a drip emitter. This is counteracted by the use of fertigation to control the concentration of minerals in the active root zone. By adding soluble fertiliser to the irrigation water, the root system can always be exposed to optimal levels of mineral nutrients at high water availabilities. Uptake by the vines is efficient due to the ability of the root system to rapidly exchange water and minerals between individual roots.

NPK concentrations in the region of 150:20:150 parts per million are ideal for application to stimulating the vegetative growth required in the early part of the growing season. These concentrations are lowered as the season progresses. Nitrogen is never added after veraison, but potassium and phosphorus can be added at low concentrations until harvest. Fertigation is ideal for the application of microelements and uptake of iron chelate is particularly efficient in these systems.

It is important to select the nutrient solution composition on the basis of the vine's needs. Petiole analysis is used for this purpose. Sampling is usually carried out at full bloom, veraison and harvest. Government advisory or extension departments in wine-growing regions publish recommended nutrient levels for each sampling season. Nitrate nitrogen levels are often unstable in petioles at full bloom, with more accurate readings found pre harvest. Potassium levels are most accurately reflected at harvest because at this time, the maximum concentrations are present in fruit and minimum in the petiole. In table grapes high potassium is desirable – grapes may be harvested with lower sugar content, because potassium increases the juice pH masking the sourness perception. Phosphorus increases sugar accumulation in the berries, but reduces their size; this is important for wine grapes but can be either undesirable or useful for table grapes depending on local conditions and varieties grown.

